



Roll No.	
Name	
Class & Section	

APEEJAY COMMON ANNUAL EXAMINATION, 2019-20

CHEMISTRY

Time Allowed : 3.00 Hrs.

Class – XI

Maximum Marks : 70

General Instructions :

- (a) *This question paper contains 37 Questions.*
- (b) *All questions are compulsory.*
- (c) *Section A contains 20 questions of one mark,
Section B contains 7 questions of two marks,
Section C contains 7 questions of three marks,
Section D contains 3 questions of five marks each.*
- (d) *Section A comprises of MCQs, Assertion-Reason Questions and Very Short Answer Questions. These questions are to be answered in one word, one sentence or as per the exact requirement of the question.*
- (e) *There is no overall choice. However, an internal choice has been provided in 2 questions of one mark, 2 questions of two marks, 4 questions of three marks and all the 3 questions of five marks.*
- (f) *Use of calculators is not permitted. You may use logarithmic tables if required.*

Section-A

Read the given passage and answer the questions 1 to 5 that follow :

The periodic table houses a few families and group of elements, each having its own particular properties. Early p-block comprises of group 13 and 14. Group 13 consists of B, Al, Ga, In, Tl with general configuration ns^2np^1 . Group 14 is Carbon family and comprises of C, Si, Ge, Sn, Pb with general configuration ns^2np^2 . Boron and carbon family form versatile compounds. Boric acid is used as an antiseptic for eyes whereas zeolites are shape selective catalysts.

1. What happens when boric acid is heated above 370K? 1
2. Name a zeolite which is used to convert alcohols directly into gasoline. 1
3. Draw the structure of Diborane. 1
4. What happens when diborane reacts with ammonia? 1
5. Why is PbCl_2 more stable than PbCl_4 ? 1

Questions 6 to 10 are one word answers :

6. 1 litre of a gas weighs 1.964 g at STP. What is its molecular mass? 1
7. Amongst ClF_3 or BCl_3 , which molecule has bond angle 120° . 1

OR

Based on VSEPR theory, total number of 90° bond angles present in the molecule of SF_4 are

8. The Vander waal's constant 'a' for different gases are given below : 1

Gas	a (atm L ⁻² mol ⁻² .)
O_2	1.36
N_2	1.39
CH_4	2.25
NH_3	4.17

Which gas can be most easily liquified?

9. Water is a liquid whereas H_2S is a gas. Give reason. 1
10. Algae reduce dissolved oxygen in water. This process is called 1

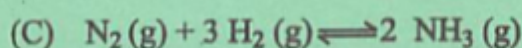
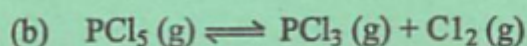
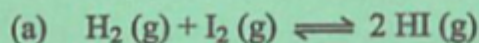
Questions 11 to 15 are multiple choice questions :

11. Which of the following pairs of d orbitals have electron density along the axis? 1
 - (a) d_{x^2} , d_{xz}
 - (b) d_{yz} , d_{xz}
 - (c) d_{x^2} , $d_{x^2-y^2}$
 - (d) d_{zy} , $d_{x^2-y^2}$
12. In second period of periodic table, first ionization enthalpy follows the order. 1
 - (a) $\text{Ne} > \text{F} > \text{O} > \text{N} > \text{C} > \text{Be} > \text{B} > \text{Li}$
 - (b) $\text{Ne} > \text{F} > \text{N} > \text{C} > \text{O} > \text{Be} > \text{B} > \text{Li}$

(c) $\text{Li} > \text{B} > \text{Be} > \text{C} > \text{O} > \text{N} > \text{F} > \text{Ne}$

(d) $\text{Ne} > \text{F} > \text{N} > \text{O} > \text{C} > \text{Be} > \text{B} > \text{Li}$

13. In which of the following reactions, the equilibrium remains unaffected on addition of small amount of argon at constant volume? 1



(d) The equilibrium will remain unaffected in all the three cases.

14. Standard electrode potential of three metals A, B and C are -1.3 V , $+0.8\text{ V}$ and -2.8 V respectively. The reducing power of these metals will be- 1

(a) $\text{A} > \text{B} > \text{C}$

(b) $\text{C} > \text{A} > \text{B}$

(c) $\text{C} > \text{B} > \text{A}$

(d) $\text{B} > \text{C} > \text{A}$

15. CO_2 is released most readily on heating which of the following? 1

(a) MgCO_3

(b) CaCO_3

(c) K_2CO_3

(d) Na_2CO_3

OR

Which of the following has lowest melting point?

(a) CaCl_2

(b) CaBr_2

(c) CaI_2

(d) CaF_2

Questions 16 to 20

(a) Both assertion and reason are correct statements and reason is the correct explanation of the assertion.

(b) Both assertion and reason are correct statements, but reason is not correct explanation of the assertion.

(c) Assertion is correct, but reason is wrong statement.

(d) Assertion is wrong, but reason is correct statement.

16. **Assertion :** Liquids tend to have maximum number of molecules at their surface. 1

Reason : Small liquid drops have spherical shape.

17. **Assertion :** One atomic mass unit is defined as one twelfth of the mass of one carbon -12 atom.

Reason : Carbon-12 isotope is the most abundant isotope of carbon and has been chosen as standard. 1

18. **Assertion :** A liquid crystallizes into a solid and is accompanied by decrease in entropy. 1

Reason : In crystals, molecules organize in an ordered manner.

19. **Assertion :** Permanent hardness of water cannot be removed by treatment with washing soda. 1

Reason : Washing soda reacts with soluble magnesium and calcium sulphate to form insoluble carbonates.

20. **Assertion :** Among isomeric pentanes, neopentane has least boiling point. 1

Reason : With an increase in branching, boiling point decreases.

Section-B

21. In the estimation of sulphur by Carius method, 0.468 g of an organic compound gives 0.668g of barium sulphate. Find the percentage of sulphur in the given compound. 2

[At mass; Ba = 137 u, S = 32 u, O = 16 u]

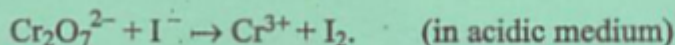
22. Calculate the energy associated with the first orbit of He^+ . What is the radius of this orbit? 2

OR

An electron has speed of 40 m sec^{-1} accurate upto 99.99%. What is uncertainty in locating its position? (Mass of electron = $9.1 \times 10^{-31} \text{ kg}$, velocity of electron = 300 ms^{-1} , $h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$)

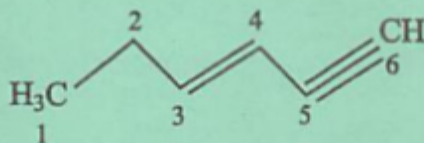
23. How is hydrogen peroxide prepared industrially? Explain why it is stored in wax-lined glass or plastic bottles? 2

24. Balance the equation : 2



25. What is BOD? If the value of BOD is low, what does it indicate? 2

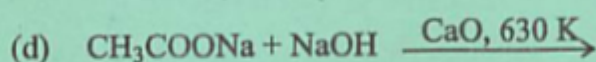
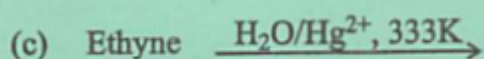
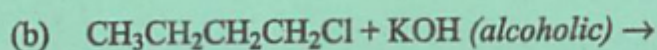
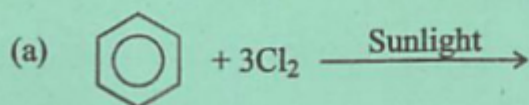
26. Write the IUPAC name of the following :



(a) Out of the 1st, 3rd and 5th carbon atom, which is most acidic? Why? 2

(b) How many sigma bonds are there in this compound?

27. Predict the products in the following reactions :



OR

Give complete balanced chemical equations for the following reactions : 2

(a) Wurtz reaction,

(b) Friedel Craft Acylation

Section-C

28. (a) Which is smaller in size- Fe²⁺ or Fe³⁺? Why? 3

(b) Write the general electronic configuration of f block elements.

(c) Write the IUPAC name and symbol of element with atomic number 117.

29. A compound on analysis found to contain following percentage composition Na = 43.4%, C = 11.3 %, and O = 45.3%. Determine the empirical and molecular formulae of this compound. The relative molecular mass of the compound is 106. (Atomic Mass of Na = 23, C = 12, O = 16) 3

OR

Commercially available Hydrochloric acid contains 38% HCl by mass.

(a) What is the molarity and molality of the solution if the density of the solution is 1.19 g/cc? 3

(b) What volume of concentrated HCl is needed to make 1 L of 0.2 M HCl solution?

30. Give the reactions involved in the Solvay's process for the preparation of sodium carbonate. 3

OR

- (a) Draw the structures of - 3
 (i) BeCl_2 (vapour) (ii) BeCl_2 (solid)

(b) Why are alkali metals strong reducing agents?

31. (a) An organic compound decomposes before attaining its boiling point. As this compound cannot be separated and purified by using simple distillation, suggest the method that can be used for its purification.

(b) What are free radicals? Discuss the stability of primary, secondary and tertiary free radicals. 3

32. (a) What will be the volume of hydrogen when 3 litres of it are cooled from 15°C to -73°C at constant pressure. 3

(b) What is Critical temperature?

OR

(a) Compare graph of P vs. V for real and an ideal gas.

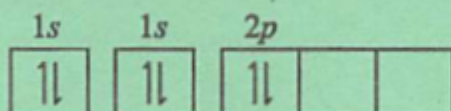
(b) What will be the pressure exerted by a mixture of 3.2g of methane and 4.4g of carbon dioxide at 9dm^3 flask at 27°C ? ($R=0.082\text{ L atm K}^{-1}\text{ mol}^{-1}$) 3

33. (a) Using s, p, d, f notations, describe the orbital with the following quantum number values :

(i) $n=4; l=0$ (ii) $n=3; l=2$

(b) Calculate de Broglie wavelength of an electron moving with 1% the speed of light.

(c) Which rule is violated in the following orbital diagram? 3



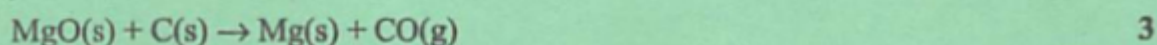
34. (a) Calculate the energy required to raise the temperature of 10 g of iron from 25°C to 500°C . (the specific heat capacity of iron is $0.45\text{ J }^\circ\text{C}^{-1}\text{ g}^{-1}$.)

(b) What mass of gold (specific heat capacity = $0.13\text{ J }^\circ\text{C}^{-1}\text{ g}^{-1}$) can be heated through the same temperature difference when supplied with the same amount of energy? 3

OR

(a) State the Third Law of Thermodynamics.

(b) Calculate and find whether it is possible to reduce MgO using carbon at 298 K.



APEEJAY SCHOOL, SAKET
FIRST TERM EXAMINATION (2018-19)
CHEMISTRY
CLASS-XI

TIME-3Hrs

M.M.-70

GENERAL INSTRUCTIONS:

- All questions are compulsory.
- Question numbers 1 to 5 are very short answer questions and carry 1 marks each.
- Questions numbers 6 to 12 are short answer questions and carry 2 marks each.
- Question numbers 13 to 24 are short answer questions and carry 3 marks each.
- Question numbers 25 to 27 are long answer questions and carry 5 marks each.
- Use log tables if necessary.

- | | | |
|----|--|---|
| 1. | Why molality is preferred over molarity in expressing the concentration of a solution? | 1 |
| 2. | Which out of Cu^{2+} , Fe^{2+} and Cr^{3+} has highest paramagnetism and why? | 1 |
| 3. | Define electronegativity. | 1 |
| 4. | Define the term bond order. | 1 |
| 5. | Gases possess characteristic critical temperatures which depend upon the magnitude of intermolecular forces between the gas particles. Critical temperature of ammonia and carbon dioxide are 405.5 K and 304.10 K. Which of these gases will liquefy first when you start cooling from 500 K to their critical temperature? | 1 |
| 6. | Commercially available concentrated hydrochloric acid contains 38% HCl by mass. What is the molarity of this solution? The density is 1.10 g/cm^3 . | 2 |
| 7. | Carbon is found to form two oxides, which contain 42.9% and 27.3% of carbon respectively. Show that these figures illustrate the law of multiple proportion. | 2 |

OR

Determine the molecular formula of an oxide of iron in which the mass percentage of iron and oxygen are 69.9 % and 30.1 % respectively, Given that the molar mass of oxide is $159.89 \text{ g mol}^{-1}$.

(Atomic mass of Fe = 55.85 g mol^{-1})

- | | | |
|-----|--|---|
| 8. | Account the following :
i) Cations are smaller and anions are larger in radii than their parent atoms.
ii) Cl has more negative electron gain enthalpy than F. | 2 |
| 9. | Explain why-
i) Be has higher ionization enthalpy than B?
ii) First electron gain enthalpy of O is negative but second is positive? | 2 |
| 10. | Draw the box structure with orbital representation of the following molecules:
i. SF_4
ii. PCl_5 | 2 |

Predict the hybridization of the central metal atom and shape of the molecule.

11. How many σ and π bonds are present in the following? 2
 - i) $\text{CH}_2=\text{C}=\text{CH}_2$
 - ii) $\text{CH}_3-\text{CH}=\text{CH}-\text{C}\equiv\text{CH}$
12. Differentiate between bonding and anti-bonding molecular orbital. 2
13. a) State Dalton's law of partial pressure. 3
 b) Derive relationship between partial pressure of a gas and its mole fraction in a mixture of two non reacting gases.

OR

- a) Explain why during fire polishing of glass the sharp edges become smooth.
 b) Derive relationship between molar mass and density of a gas.
14. a) Discuss the shape of BrF_3 on the basis of VSEPR theory. 3
 b) On the basis of Molecular orbital theory explain, why is there an increase in bond order in going from O_2 to O_2^+ while a decrease in going from N_2 to N_2^+ ?
15. 3 g of H_2 reacts with 29g of O_2 to form H_2O . 3
 - a) Which is the limiting reagent?
 - b) Calculate the maximum amount of H_2O that can be formed.
 - c) Calculate the amount of the reactant that is left unreacted.
16. a) Calculate the mole fraction of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) and water in a solution containing 20% of $\text{C}_2\text{H}_6\text{O}_2$ by mass. 3
 b) What is Gay Lussacs's law of gaseous volumes?
17. i) State and explain Heisenberg's uncertainty principle. 3
 ii) A golf ball has mass 40 g and speed 45 m s^{-1} . If the speed can be measured with accuracy of 2%. Calculate the uncertainty in position of the golf ball.
18. An atom has 2K, 8L and 3M electrons. Write down its electronic configurations and indicate in it. 3
 - i) no. of subshells
 - ii) no. of orbitals
 - iii) no. of s electrons
 - iv) no. of unpaired electrons.
19. Amongst the elements of the third period (Na to Ar), Identify the element 3
 - i. with highest ionization enthalpy
 - ii. with largest atomic radii
 - iii. most reactive non metal
 - iv. most reactive metal
 - v. an element that shows characteristic properties of metals as well as non-metal.
 - vi. an element whose oxide is amphoteric in nature.
20. Give reasons: 3
 - i) Successive ionisation enthalpies are higher.
 - ii) Halogens have the highest negative electron gain enthalpy.
 - iii) Electron gain enthalpy values of noble gases are positive while those of Be, Mg, N and P are almost zero.
21. i) Differentiate between Schottky & Frenkel defects. 3
 ii) Calculate the packing efficiency of a face centred cubic unit cell.
22. a) Write the half cell reaction and the overall cell reaction for the electrochemical cell 3



Calculate the standard E.M.F. for the cell if the standard electrode potentials (reduction) for Pb^{2+}/Pb and Zn^{2+}/Zn electrodes are -0.126V and -0.763V respectively.

23. a) Give two main functions of the salt bridge in an electrochemical cell. 3
 a) Assign oxidation numbers to the underlined elements in each of the following species:
 (i) $\text{Na}_3\underline{\text{P}}\text{O}_4$
 (ii) $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$

b) Balance the following redox reactions:

- $\text{MnO}_4^- (\text{aq}) + \text{Br}^- (\text{aq}) \rightarrow \text{MnO}_2 (\text{s}) + \text{BrO}_3^- (\text{aq})$ (In basic medium)
 24. i) What is meant by PCBs? 3
 ii) What is SMOG? How is classical smog different from photochemical smog?
 25. (i) Explain Hund's rule of maximum multiplicity with an example. 5
 (ii) The work function for Caesium atom is 1.9 eV . Calculate (a) threshold wavelength and (b) the threshold frequency of the radiation. If the Caesium element is irradiated with a wavelength 500 nm , calculate the kinetic energy of the ejected photoelectron.

OR

- (i) Explain-
 (a) Photoelectric effect
 (b) Pauli's exclusion principle
 (ii) Using s, p, d, f notations, describe the orbital with the following quantum number
 (a) $n = 4, l = 0$, (b) $n = 3, l = 2$
 (ii) Emission transition in the Paschen series ends at orbit $n=3$ and start from an orbit n and can be represented as $\nu = 3.29 \times 10^{15} (1/3^2 - 1/n^2)\text{ Hz}$. Calculate the value of n if its transition is obtained at 1285 nm .
 26. (a) Describe in brief how sp, sp^2, sp^3 hybrid orbitals are formed using suitable examples. 5
 (b) What do you mean by hydrogen bond? Explain with suitable example. What is the difference between intermolecular and intramolecular hydrogen bonding?

OR

- (a) Distinguish between a sigma (σ) and a pi (π) bond.
 (b) Use the molecular orbital theory to explain why Be_2 molecule does not exist.
 (c) Define resonance with a suitable example.
 27. a) Calculate the total pressure in a mixture of 8 g of dioxygen and 4 g of dihydrogen confined in a vessel of 1 dm^3 at 27°C . ($R = 0.083\text{ bar dm}^3\text{ K}^{-1}\text{ mol}^{-1}$) 5
 b) Explain the following :-
 i. Ferromagnetic substances
 ii. Ferrimagnetic substances



ROLL NO.	
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APEEJAY COMMON ANNUAL EXAMINATION, 2018-19

04

CLASS-XI
CHEMISTRY

Time allowed : 3 hrs.

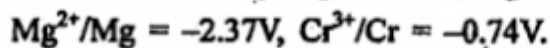
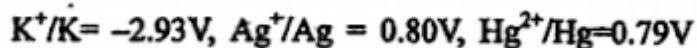
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- (v) Q. Nos. 13 to 24 are short answer questions of three marks each. Answer these in about 40 words each.
- (vi) Q. Nos. 25 to 27 are long answer questions of five marks each. Answer these in about 70 words each.
- (v) Use log tables if necessary. Calculators are not permitted.

1. Calculate the amount of water(g) produced by the combustion of 16g of methane ? (atomic mass of C=12, H=1, O=16). (1)

2. Given the standard electrode potentials,



Arrange these metals in their increasing order of reducing power.

OR

Calculate the oxidation number of Mn in K_2MnO_4 and N in HNO_3 (1)

3. Why is LiI more soluble than KI in ethanol ? (1)

P.T.O.



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APEEJAY COMMON ANNUAL EXAMINATION, 2018-19

04

CLASS-XI
CHEMISTRY

Time allowed : 3 hrs.

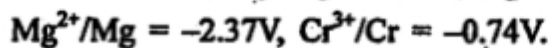
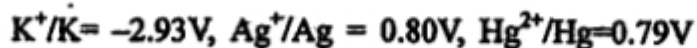
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P.T.O.

CBSE Question Paper 2019 (Set-2)

Class 11 Chemistry

MPVM Gangagurukulam

Duration: 3 hrs.

M.Marks: 70

General Instructions:

- i. Q1 - 5 are of 1 marks each.
- ii. Q6 - 10 are of 2 marks each.
- iii. Q11-23 are of 3 marks each.
- iv. Q24 is of 4 marks.
- v. Q25 - 27 are of 5 marks each.

1. BF_3 does not have proton but still acts as an acid. Why?
2. Write the IUPAC name of the following compound: $\text{Br}-\text{CH}_2-\text{CH}=\text{CH}-\text{COOH}$.
3. Arrange the following elements in the increasing order of electronegativity Si, C, P and N.
4. Give one example of disproportionation reaction.
5. What is the slope between PV and P at constant temperature?
6. What will be the molality of the solution containing 18.25 g of HCl gas in 500 g of water?
7. On the basis of VSEPR theory, explain the shape of ClF_6 and SF_4 .
8. Complete the following reactions:
 - i. $2\text{CH}_3\text{Br} + 2\text{Na} \xrightarrow{\text{dry ether}}$
 - ii. $\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow{\text{alc. KOH}}$
9. The equilibrium constant K_p for the thermal dissociation of PCl_5 at 200°C is 1.6 atm. What is the pressure at which it is 50% dissociated?
10. Write short notes on:
 - i. Acid rain
 - ii. Green house effect
11. The density of certain gaseous oxide at 1.5 bar pressure at 10°C is same as that of dioxygen at 20°C & 4.5 bar pressure. Calculate the molar mass of gaseous oxide.

12. What causes the temporary & permanent hardness of water? Explain one method for the removal of permanent hardness.
13. In an organic compound C, H & N are present in the 9: 1 : 3.5 ratio by weight if molecular weight of the compound is 108, then calculate the molecular formula of the compound.
14. How would you account for the following:
 - i. $\text{Ti}(\text{NO}_2)_3$ acts as an oxidizing agent.
 - ii. Although B - Cl has a dipole moment but still BCl_3 is a non - polar molecule
 - iii. Complete the following reaction:

$$\text{PbO}_3 + \text{NaOH} \rightarrow$$
15.
 - i. Why is NF_3 pyramidal but BCl_3 is a non - polar molecule
 - ii. Which d - orbital is involved in sp^3d hybridisation?
 - iii. Solubility of MgCl_2 much greater than MgF_2 , Why?
16. Write balanced chemical equation for the reaction of potassium permanganate with oxalic acid & potassium dichromate with potassium iodide in acidic medium.
17. Explain why:
 - i. Electron affinity is positive when O changes into O^{2-} .
 - ii. Be has higher first I.E than B.
 - iii. Cl has more negative electron gain enthalpy than F.
18.
 - i. The electron configuration of a dipositive ion M^{2+} is 2, 8, 14 & it's mass number is 56. How many neutrons are present in It?
 - ii. What Is the most probable radius for finding the electron in He^+
 - iii. What is the shape of the orbital with the value of $l = 2$ & $m = 0$?
19.
 - i. What is the hybridisation of each carbon in $\text{H}_2\text{C}=\text{C}=\text{CH}_2$?
 - ii. Show the polarization of carbon - magnesium bond in the following structure:

$$\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-MgX}$$
 - iii. Draw the possible resonance structures for $\text{CH}_3 - \ddot{\text{O}} - \text{CH} = \text{CH}_3$
20. Explain the following terms:
 - i. Hund's rule
 - ii. Heisenberg's Uncertainty Principle.
 - iii. Quantum number
21. The combustion of 1 mole of benzene takes place at 298 K and 1 atm. After combustion

$\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are formed and 3267 kJ of heat is liberated. Calculate standard enthalpy of formation of benzene. Given $\Delta_f H^\circ$ of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ as - 393.5 kJ/mol and - 285.83 kJ/mol respectively.

22. Explain the following
- Solubility product
 - Lewis acid
 - Buffer solution
23. Rahul went to market to buy fruits and vegetable. The vendor put the fruits and vegetables in the polythene bag but Rahul asks the vendor to put the things In the Jute beg which he had carried with him.

Answer the following Questions:

- Why did Rahul refused to use polythene bag?
 - As a chemistry student, why would you advocate the use of jute bags instead of polythene bags?
 - What are the values possessed by Rahul?
 - Suggest two activities to ban the use of polythene bags.
24. What are reaction intermediates? Explain the stability of carbocations and carbanions on the basis of inductive effect.
25. i. Explain why
- Boron forms electron deficient compounds.
 - Aluminium chloride exists as a dimer.
 - BBr_3 is a stronger acid than BF_3 .
- ii. Explain the structure of diborane.
26. i. Explain the mechanism of Friedal crafts Alkylation reaction.
- ii. Explain the following reactions with examples: (a) Wurtz reaction (b) Nitration of Benzene (c) Ozonolysis of Alkenes.

CBSE Question Paper 2019 (Set-2)

Class 11 Chemistry

MPVM Gangagurukulam

Duration: 3 hrs.

M.Marks: 70

General Instructions:

- Q1 - 5 are of 1 marks each.
 - Q6 - 10 are of 2 marks each.
 - Q11-23 are of 3 marks each.
 - Q24 is of 4 marks.
 - Q25 - 27 are of 5 marks each.
- BF_3 does not have proton but still acts as an acid. Why?
 - Write the IUPAC name of the following compound: $\text{Br}-\text{CH}_2-\text{CH}=\text{CH}-\text{COOH}$.
 - Arrange the following elements in the increasing order of electronegativity Si, C, P and N.
 - Give one example of disproportionation reaction.
 - What is the slope between PV and P at constant temperature?
 - What will be the molality of the solution containing 18.25 g of HCl gas in 500 g of water?
 - On the basis of VSEPR theory, explain the shape of ClF_6 and SF_4 .
 - Complete the following reactions:
 - $2\text{CH}_3\text{Br} + 2\text{Na} \xrightarrow{\text{dry ether}}$
 - $\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow{\text{alc. KOH}}$
 - The equilibrium constant K_p for the thermal dissociation of PCl_5 at 200°C is 1.6 atm. What is the pressure at which it is 50% dissociated?
 - Write short notes on:
 - Acid rain
 - Green house effect
 - The density of certain gaseous oxide at 1.5 bar pressure at 10°C is same as that of dioxygen at 20°C & 4.5 bar pressure. Calculate the molar mass of gaseous oxide.

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Session Ending Examination (2015-16)

Chemistry

CBSE CLASS – XI

Time: 3 Hrs. M.M: 70

General Instructions:

- (i) Question 1 to 5 one mark.
- (ii) Question 6 to 10 each two mark.
- (iii) Question 11 to 22 each three mark.
- (iv) Question 23 is value based question and carry four marks.
- (v) Questions 24 to 26 each five mark.

Section A

- 1. Which two elements among the following belong to the same period Al, Si, Ba and O?
- 2. Why liquids diffuse slowly as compared to gases?
- 3. A gas is filled into a bulb connected to an open limb manometer. The level of mercury in the open arm is 2.1 cm lower than that in the other arm of manometer. The atmospheric pressure is 720 mm. What is the pressure of the gas?
- 4. Heat capacity is an extensive property but specific heat is intensive property. Comment on the statement.
- 5. Boron forms no compounds in unipositive state but thallium in unipositive state is quite stable. Explain.

Section B

- 6. Cyanogen (CN), is known as a pseudohalogen because it has some properties similar to halogens. It is composed of two CN's joined together. Do the two CN's join through the carbon or the nitrogen (i. e., C – N – N – C or N – C – C – N)?
- 7. CO_2 is heavier than N_2 and O_2 gases present in the air but it does not form the lower layer of the atmosphere. Explain.
- 8. Calculate
 - (i) mass of 2.5 g atoms of magnesium,
 - (ii) gram atom in 1.4 g of nitrogen (Atomic mass Mg = 24, N = 14)

Or

Calculate the mass of

(i) 1 molecule of N_2 ,

(ii) 100 molecules of sucrose ($C_{12}H_{22}O_{11}$).

9. What happens when (give chemical equation only)

(i) sodium is exposed to moist air?

(ii) sodium reacts with water?

10. Name one reaction in which water acts

(i) as an oxidising agent and

(ii) as a reducing agent?

Section C

11. The threshold wavelength for emitting photons from a metal is $6.0 \times 10^3 \text{ \AA}$. What would be the wavelength of radiation to produce photoelectrons having twice the kinetic energy of those produced by radiation of wavelength $3 \times 10^3 \text{ \AA}$?

12. The wavelength of the first line in the Balmer series is 656 nm. Calculate the wavelength of the second line and the limiting line in Balmer series.

13. A student while measuring pressure and volume of a gas at constant temperature forgot to record some of the observations as follows.

Pressure (torr)	100	125	200	(-)
Volume (mL)	80	(-)	40	32

Calculate the missing observations. Name the law used in your calculations.

14. Give the balanced equations for the reactions that form the basis for the following redox titrations

(i) Titration of potassium permanganate for estimation of ferrous ions.

(ii) Titration of iodine against sodium thiosulphate (iodimetric titration).

(iii) Iodometric titration for estimation of cupric ions.

15. The degree of dissociation of N_2O_4 into NO_2 at one atmospheric pressure and 313 K is 0.310. Calculate K_p of the dissociation reaction at this temperature. What will be the degree of dissociation at 10 atmospheric pressure at the same temperature?

Or

When 3.06 g of solid NH_4HS is introduced into a two-litre evacuated flask at 27°C , 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide.

(i) Calculate K_c and K_p for the reaction at 27°C .

(ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?

16. What do you understand by hydrogen economy?

17. An organic acid contains C, H and O. A 4.21 mg sample of acid is completely burned. It gives 6.21 mg of carbon dioxide and 2.54 mg of water. What is the mass percentage of each element in the acid?

18. How does BF_3 react with

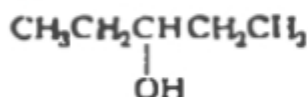
(i) ammonia

(ii) lithiumhydride

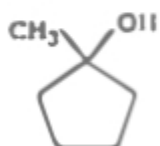
(iii) water?

19. Give the structure and IUPAC name of the alkenes formed from acid-catalysed dehydration of

(i)



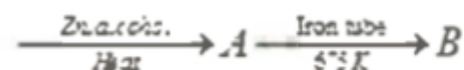
(ii)



(iii)



20. (i) 1, 1, 2, 2, -tetrachloroethane



(ii) Acetylene $\xrightarrow{\text{NaNH}_2}$ A $\xrightarrow{\text{CH}_3\text{CH}_2\text{Br}}$ B

(iii) Propyne $\xrightarrow[\text{Quinoline}]{\text{H}_2, \text{Pd, BaSO}_4}$ A $\xrightarrow[\text{(ii) Zn, H}_2\text{O}]{\text{(i) O}_3}$ B

21. (i) Why PbCl_4 is less stable than SnCl_4 ?

(ii) What happens when silicon dioxide is treated with hydrogen fluoride? (Give reactions)

only).

(iii) Why is $PbCl_4$ good oxidising agent?

22. Calculate the pH of a solution formed by mixing 0.2 M NH_4Cl and 0.1 M NH_3 . The pK_b of ammonia is 4.75.

Section C

23. Anil and Subha are science students. They want to prepare 51 g of NH_3 by combining dinitrogen and dihydrogen. They have 153 g of hydrogen. According to Anil's view for this, they have to mix 25.5 kg of nitrogen to the hydrogen while Subha suggestion was that they have to mix 42 g of nitrogen to the hydrogen.

(i) According to you whose suggestion was correct and why?

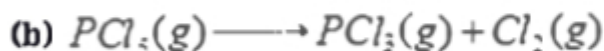
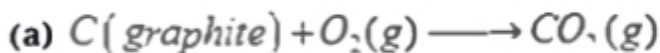
(ii) Which is the limiting reagent here?

(iii) Find the excess reagent.

(iv) Comment on the values associated with Anil and Subha?

Section D

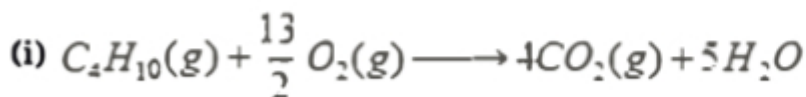
24. (i) Predict whether $\Delta H > \Delta U$ or $\Delta H < \Delta U$ or $\Delta H = \Delta U$ for



(ii) Determine the value of ΔH and ΔU for the reversible isothermal evaporation of 90.0 g of water at $100^\circ C$. Assume that water vapour behaves as an ideal gas and heat of evaporation of water is 540 cal g^{-1} . ($R = 2.0 \text{ cal mol}^{-1} K^{-1}$).

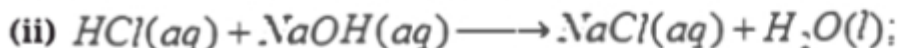
Or

Fill in the blanks:



$$\Delta H = -2878 \text{ kJ}$$

ΔH is the enthalpy of of butane gas.



$$\Delta H = -57.1 \text{ kJ in real terms,}$$



Session Ending Examination (2015-16)

Chemistry

CBSE Class – XI

Time: 3 Hrs. M.M: 70

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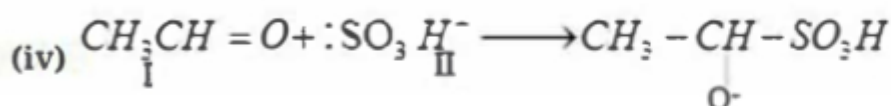
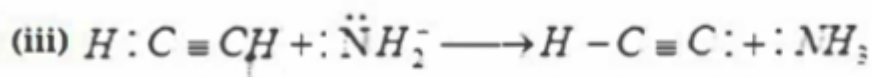
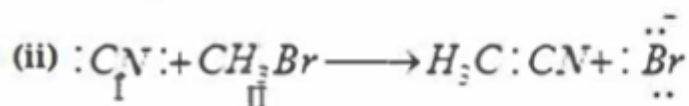
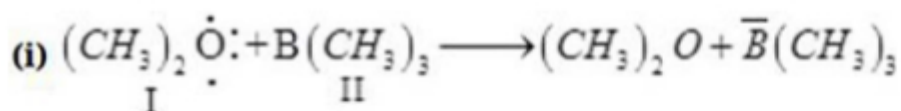
Section A

1. When α -rays hit a thin foil of gold, very few α -particles are deflected back. What does it prove?
2. Calculate the number of atoms of each type in 5.3 g of Na_2CO_3 .
3. Critical temperatures of ammonia and carbon dioxide are 405.5 K and 304.10 K respectively. Which of these gases will liquefy first when you start cooling from 500 K to their critical temperature?
4. Why does table salt get wet in rainy season?
5. What orbitals can overlap to form a σ – bond and which orbitals can do so to form a π – bond?

Section B

6. What is hybrid state of central atom in the following? NO_3^- , BF_4^- , PF_5 , IF_5 and CO_2 .
7. Using van der Waals equation, calculate the constant 'a' when two moles of a gas confined in a four litre flask exerts a pressure of 11.0 atmosphere at a temperature of 300 K. The value of 'b' is 0.05 L mol^{-1} .
8. (i) What are conjugate acid-base pairs?
(ii) What are amphoteric substance? Give one example.
9. What are allotropes? List two characteristic differences between diamond and graphite which are allotropes of carbon.

10. Identify the species I and II as electrophiles (E) or nucleophiles (Nu) in the reactions



Section C

11. Arrange the elements of second row in order of increasing second ionisation enthalpies.

12. 0.145 dm^3 of hydrogen gas is collected over water at 23°C and total pressure of 99085 Nm^{-2} . If the vapour pressure of water at 23°C is 2973 Nm^{-2} , what is the pressure exerted by the dry gas? Also calculate the volume of dry gas? Under NTP conditions?

13. What volume of 0.6 M HCl has enough hydrochloric acid to react exactly with 25 mL of aqueous NaOH having concentration of 0.5 M?

14. When a certain metal was irradiated with a light of frequency $3.2 \times 10^{16} \text{ Hz}$, the photoelectrons had twice the kinetic energy as emitted when the same metal was irradiated with light of frequency $2.0 \times 10^{16} \text{ Hz}$. Calculate the threshold frequency (ν_0) of the metal.

15. Calculate the kinetic energy of moving electron which has a wavelength of 4.8 pm (mass of electron = $9.11 \times 10^{-31} \text{ kg}$, $h = 6.63 \times 10^{-34} \text{ Js}$).

16. Explain the difference in the reactivity of the group, metals with water.

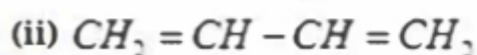
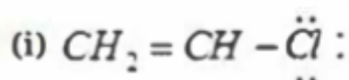
Or

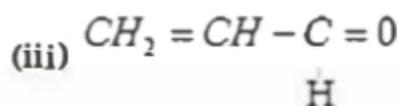
(i) What is the oxidation state of K in KO_2 ?

(ii) LiH is more stable than NaH. Explain.

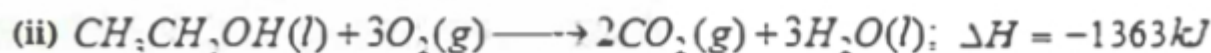
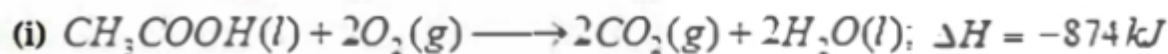
(iii) Why are alkali metal halides soluble in water?'

17. Draw the resonance structure of the following compounds





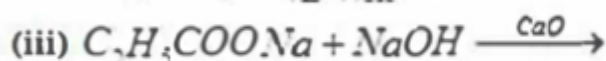
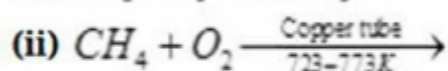
18. The enthalpy changes for the following reactions at 298 K and 1 atm are given below:



(iii) Calculate the internal energy changes for these reactions.

19. Two liquids (A) and (B) can be separated by the method of fractional distillation. The boiling point of liquid (A) is less than boiling point of liquid (B). Which of the liquids do you expect to come out first in the distillate? Explain.

20. Complete the reactions,



21. Identify the kind of energy change usually associated with each of the following.

(i) Automobile engine

(ii) Fluorescent lamp

(iii) Radio

(iv) Friction

(v) Photoelectric cell

22. How should the management of domestic waste be done?

Section D

23. WHO has suggested that people should take less common salt as it leads to high blood pressure and heart ailments. Sodium ions are found primarily on outside of the cells, being located in blood plasma and in the interstitial fluid which surrounds the cells. These ions participate in the transmission of nerve signals, in regulating the flow of water across cell membranes and in the transport of sugar and amino acids into cells.

(i) How much sodium is present in blood plasma?

(ii) What is the major source of sodium?

(iii) What is the harmful effect if we take excess of salt?

(iv) What values are possessed by WHO?

Session Ending Examination (2015-16)

Chemistry

CBSE Class – XI

Time: 3 Hrs. M.M: 70

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Section A

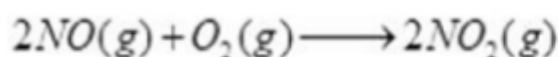
- 1. Arrange the following in the increasing order of their size F^- , Li^+ , Na^+ , Cl^- .
- 2. Why are liquids like ether and acetone kept in cool places?
- 3. Why do trihalides of group 13 elements fume in the moist air?
- 4. What experimental support is available for de-Broglie's equation?
- 5. Why is second electron gain enthalpy (electron affinity) negative (i.e., energy is absorbed)?

Section B

- 6. Calculate the uncertainty in the velocity of a wagon of mass 3000 kg whose position is known to an accuracy of ± 10 pm. (Planck's constant = $6.63 \times 10^{-34} \text{ Js}$).
- 7. Draw the possible resonance structures for $CH_3 - \ddot{O} - \overset{+}{C}H_2$ and predict which of the structures is more stable. Give reason for your answer.
- 8. On Illustrate the formation of bonding and antibonding molecular orbitals from atomic orbitals?
- 9. With the help of Le-Chatelier 's principle explain
 - (i) we sweat more on a humid day.
 - (ii) clothes dry quickly on a windy day.
- 10. What is the role of red phosphorus in the reduction of alkyl halide using hydriodic acid?

Section C

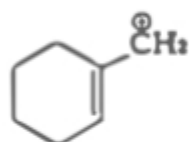
11. Calculate the number of moles of nitrogen dioxide, NO_2 that could be prepared from 0.35 mole of nitrogen oxide and 0.25 mole of oxygen.



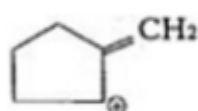
Identify the limiting reagent and the excess reagent in the reaction. What would happen to the potential yield of NO_2 if the amount of NO were increased? What, if the amount of O_2 were increased?

12. (i) Which of the following ions is more stable?

Use resonance to explain your answer.

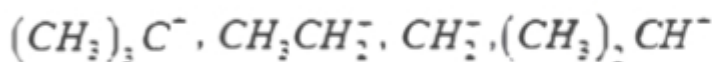


(A)



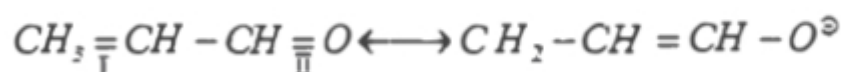
(B)

(ii) What is the correct order of stability of the carbonium ion?



Or

Resonance structures of propenal are given below. Which of these resonating structures is more stable? Give reason for your answer.



13. Draw the Lewis structures for the following ionic compounds.

(i) Li_2O

(ii) CaCl_2

(iii) LiF

14. The enthalpy change involved in the oxidation of glucose is $-2880 \text{ kJ mol}^{-1}$. 25% of this energy is available for muscular work. If 100 kJ of muscular work is needed to walk one kilometer, what is the maximum distance that a person will be able to walk after Eating 120 g of glucose?

15. 0.1 mole of $\text{N}_2\text{O}_4(\text{g})$ was sealed in a tube under atmospheric conditions at 25°C

Calculate the number of moles of $\text{NO}_2(\text{g})$ present if the equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is

reached after sometime ($K_p = 0.14$).

Or

Consider the reaction



Indicate the direction in which the equilibrium will shift when

(i) temperature is increased

(ii) volume is increased

(iii) a catalyst is added

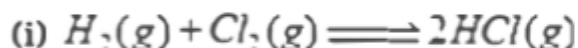
(iv) helium gas is added at constant volume so that the total pressure is increased

(v) helium gas is added at constant pressure.

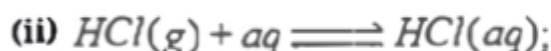
16. Calculate the molar solubility of $Ni(OH)_2$ in 0.10 M NaOH. The solubility product of

$Ni(OH)_2$, is 2.0×10^{-15} .

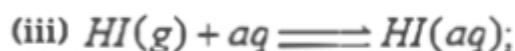
17. Calculate the enthalpy of formation of $HI(g)$ from the following data



$$\therefore \Delta_r H^\circ = -184.6 kJ$$



$$\Delta_r H^\circ = -72.43 kJ$$



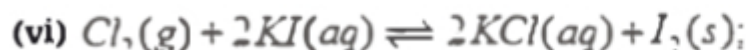
$$\Delta_r H^\circ = -80.37 kJ$$



$$\Delta_r H^\circ = -57.49 kJ$$

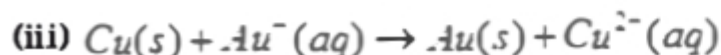
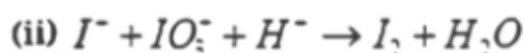
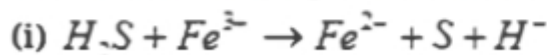


$$\Delta_r H^\circ = -57.19 kJ$$



$$\Delta_r H^\circ = -219.4 kJ$$

18. Write the half equations for the following reactions.



Session Ending Examination (2015-16)

Chemistry

CLASS – XI

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Section A

1. In terms of ionisation enthalpy and electron gain enthalpy, what type of atoms combine to form an ionic compound?
2. For testing halogens in an organic compound with $AgNO_3$ solution, sodium extract (Lassaigne's test) is acidified with dilute HNO_3 . What will happen if a student acidifies the extract with dilute H_2SO_4 in place of dilute HNO_3 ?
3. What are the shortcomings of the Wurtz reaction?
4. What is the nature of oxides formed by most of p-block elements?
5. If atomic weight of C and S are 12 and 32 respectively, then an atom of S is times heavier than an atom of C.

Section B

6. Out of N_2 and NH_3 , which one will have greater value for van der Waals' constant 'a' and 'b'?
7. If the energy difference between the electronic states of hydrogen atom is $214.68 \text{ kJ mol}^{-1}$, what will be the frequency of light emitted when the electron drops from the higher to the lower energy state?
(Planck's constant = $39.79 \times 10^{-14} \text{ kJ s mol}^{-1}$)

8. (i) Why is the first ionisation enthalpy of the transition elements reasonably constant?
 (ii) Explain the trend in ionisation enthalpies for the noble gases.
9. Write the structure of the product of hydrolysis of $(CH_3)_2SiCl_2$ and its two uses.
10. A liquid with high boiling point decomposes on simple distillation but it can be steam distilled for its purification. Explain how is it possible?

Section C

11. What is an electrovalent bond (or ionic bond)? Explain its formation with two suitable examples.
12. Calculate the uncertainty in the position of an electron if uncertainty in its velocity is
 (i) 0.001%
 (ii) zero.
 (The mass of electron = $9.1 \times 10^{-31} \text{ kg}$, velocity of electron = 300 ms^{-1}).
13. One mole of CO_2 occupies 1.5 L at $25^\circ C$. Calculate the pressure exerted by the gas using
 (i) ideal gas equation.
 (ii) van der Waals' gas equation with $a = 3.6 \text{ L}^2 \text{ bar mol}^{-2}$ and $b = 0.04 \text{ L mol}^{-1}$.
14. Arrange the following molecules in the decreasing order of oxidation state (+ve to -ve) of nitrogen NO_2 , NH_3 , HN_3 , NO_2^- , N_2H_4
15. Write three general characteristics of the elements of s-block of the periodic table which distinguish them from the elements of the other blocks.
16. To what type of system the following belong?
 (i) Tree
 (ii) Pond
 (iii) Animals
 (iv) Tea placed in kettle
 (v) Tea placed in thermos flask
 (vi) Tea placed in a cup.
17. The element 119 has not been discovered. What would be the IUPAC name and symbol for this element? On the basis of the periodic table, predict the electronic configuration of this element and the formula of its most stable chloride and oxide.
18. What is the cause of acid rain? How is it harmful to the environment?

19. (i) Calculate the energy needed to raise the temperature of 10.0 g of iron from 25°C to 250°C if specific heat capacity of iron is $0.45\text{JC}^{-1}\text{g}^{-1}$.
- (ii) What mass of copper (specific heat capacity = $0.385\text{JC}^{-1}\text{g}^{-1}$) can be heated through the same temperature difference when supplied with the same amount of energy as above.
20. Write IUPAC names of the following compounds and calculate the number of sigma (σ) and π (pi) bonds in each.
- (i) $(\text{CH}_3)_2\text{CH}-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}$
 $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}$
 C_2H_5
- (ii) 
- (iii) $\text{CH}_2=\text{C}(\text{CH}_2\text{CH}_2\text{CH}_3)_2$
- (iv) $\text{CH}_3(\text{CH}_2)_3-\text{CH}-\text{CH}=\text{C}(\text{CH}_2\text{CH}(\text{CH}_3)_2)_2$
 CH_3 C_2H_5
21. Discuss the structures of CO , CO_2 and SiO_2 ?
22. Explain the following properties of alkali metals
- (i) Conductivity (ii) Reducing power (iii) Tendency to form monovalent cation (iv) Solubility of water

Section D

23. Mrs. Kiran Yadav in the family way. Pregnant ladies and breast feeding ladies need extra calcium. They need 1300 to 1500 mg of calcium per day. Deficiency of calcium is found in ladies more than gents. The beating of heart, hormonal system, muscular coordination, working of brain and coagulation of blood need calcium, it plays an important role in nervous system. Mrs. Kiran is taking milk every day. Doctor has given her calcium supplement to fulfil her daily need.
- (i) Is it true that calcium gets stored in our body as bone bank? Is it useful in old age?
- (ii) If we do not take calcium in our diet, how does body get calcium?
- (iii) What are the consequences of the above situation?
- (iv) How is the deficiency of calcium detected?
- (v) How does baby get calcium?

Section E

Session Ending Examination (2015-16)

Chemistry

CLASS – XI

Time: 3 Hrs. M.M: 70

General Instructions:

- (i) Question 1 to 5 one mark.
- (ii) Question 6 to 10 each two mark.
- (iii) Question 11 to 22 each three mark.
- (iv) Question 23 is value based question and carry four marks.
- (v) Questions 24 to 26 each five mark.

Section A

1. Is the law of constant composition true for all types of compounds? Mention why or why not?
2. What was the necessity for the classification of the elements?
3. Water decomposes by absorbing 286.2 kJ of electrical energy per mole. When H_2 and O_2 combine to form one mole of H_2O , 286.2 kJ of heat is produced. Which law is proved? What statement of the law follows from it?
4. Which d-orbital does not have four lobes and what is its shape called?
5. Which species is the smallest aromatic substance?

Section B

6. Give two reactions to show acidic character of alkynes?
7. Calculate the mass of
 - (i) an atom of silver (atomic mass = 108).
 - (ii) 1 molecule of naphthalene ($C_{10}H_8$).
8. (i) What did Thomson's experiment on the deflection of cathode rays yield?
(ii) What experiment neglected Thomson's model of the atom as an intimate mixture of

negative and positive part?

9. Rank the following in the order of increasing entropy.

(i) 1 mole of $H_2O(l)$ at $25^\circ C$ and 1 atm pressure.

(ii) 2 moles of $H_2O(s)$ at $0^\circ C$ and 1 atm pressure.

(iii) 1 mole of $H_2O(g)$ at $100^\circ C$ and 1 atm pressure.

(iv) 1 mole of $H_2O(l)$ at $0^\circ C$ and 1 atm pressure.

10. Radius of the fourth orbit in hydrogen atom is 0.85 nm. Calculate the velocity of the electron in this orbit (mass of electron $9.1 \times 10^{-31} \text{ kg}$).

Section C

11. Calculate the wavelength of the spectral line obtained in the spectrum of Li^{2+} ion when transition takes place between two levels whose difference is 2 and the sum is 4?

12. Calculate the oxidation number of

(i) N in NO_3^-

(ii) P in $H_3P_2O_7^-$

(iii) C in CO_3^{2-}

(iv) O in OF_2

(v) Cl in ClO_4

(vi) Cr in $Cr_2O_7^{2-}$

13. (i) Name the classes of hydrides to which H_2O , B_2H_6 , CrH and NaH belong.

(ii) What do you mean by hydride gap?

14. Calculate the work of expansion when 100 g of water is electrolysed at a constant pressure of 1 atm and temperature of $25^\circ C$

15. Standard enthalpy of vaporisation of benzene at its boiling point is 30.8 kJ mol^{-1} . For how long would a 100 W electric heater have to operate in order to vaporise a 100 g sample of benzene at its boiling temperature (power = $\frac{\text{energy}}{\text{time}}$, $1 \text{ W} = 1 \text{ Js}^{-1}$)?

16. Discuss the various reactions that occur in the Solvay process.

17. Write three physical and three chemical properties of carbon monoxide.

18. What are cycloalkanes? Give the name, structural and the condensed formula of the first

four members of the cycloalkane series.

19. One mole of a hydrocarbon (A) reacts with one mole of bromine giving a dibromo compound, $C_{10}H_5Br_2$. Substance (A) on treatment with cold dilute alkaline $KMnO_4$ solution forms a compound $C_5H_{12}O_2$. On ozonolysis (A) gives equimolar quantities of propanone and ethanal. Deduce the structural formula of (A).

20. Write chemical equations only involved in the preparation of each of the following.

(i) Plaster of Paris

(ii) Quicklime

(iii) Slaked lime

Also write any one use of each.

Or

Describe the importance of the following.

(i) Limestone

(ii) Cement

(iii) Borax

21. (i) Emeralds are gem-quality forms of the mineral beryl, $Be_3Al_2(SiO_3)_6$. Calculate the percentage by mass of silicon in beryl.

(ii) The oxygen-carrying protein, known as haemoglobin, have 0.0335% Fe by mass and contains four Fe atoms per haemoglobin molecule. Calculate the molecular weight of this protein.

22. Calculate the number of moles of hydrogen gas present in 500 cm³ of the gas taken at 300 K and 760 mm pressure. If this sample of hydrogen is found to have a mass equal to 4.09×10^{-2} g, calculate the molar mass of hydrogen.

Section D

23. Syn gas is major source for the production of dihydrogen. Earlier it was obtained only by coal gasification. But scientists of many developed countries now a days obtained it from biomass gasification.

(i) Write the main difference between coal gasification and biomass gasification.

(ii) Why is biomass gasification is more advantageous over coal gasification for the

Please check total printed pages before start:- 8

Roll No. : 01

SESSION ENDING EXAMINATION, 2014-15

SUBJECT - CHEMISTRY

CLASS - XI

Time : 3:00 Hrs.

M.M. 70 MARKS

GENERAL INSTRUCTIONS :

1. All questions are compulsory.
2. Question no. 1 to 5 are very short answer questions and carry 1 mark each.
3. Question no. 6 to 10 are short answer questions and carry 2 marks each.
4. Question no. 11 to 22 are short answer questions and carry 3 marks each.
5. Question no. 23 is value based question and carry 4 marks.
6. Question no. 24 to 26 are long answer type questions and carry 5 marks each.
7. Use log table if necessary , use of calculator is not allowed.

[P.T.O.]

Q.1: What do you understand by "Limiting reagent". (1)

Q.2: State Hiesenberg Uncertainty Principle. (1)

Q.3: The value of Vander waal's constants a and b are as given for two gases :

Gases	a (Atm L ² mol ⁻²)	b (L mol ⁻¹)
CO ₂	3.6	0.043
SO ₂	6.7	0.056

Out of these two gases which gas molecules will possess largest magnitude of intermolecular forces of attraction ? (1)

Q.4: In Lassaigne's test for detection of nitrogen in an organic compound , the blue colour appears due to the formation of (1)

Q.5: For a reaction both enthalpy change and entropy change are positive. Under what conditions the reaction will be spontaneous ? (1)

Q.6: (i) How many subshells are associated with n=4 ?

(ii) How many electrons will be present in the subshells having m_s value of -1/2 for n=4 ? (1+1)

Q.7: (i) Define electron gain enthalpy .

(ii) Why is the electron gain enthalpy of chlorine more negative than fluorine ? (1+1)

Q.8: (i) In what group of the periodic table the element will found having electronic configuration : [Xe] 4f¹⁴ 5d⁴ 6s²

(ii) Why first ionization enthalpy of nitrogen is more than oxygen? (1+1)

Q.9: Which out of NH₃ and NF₃ has higher dipole moment and why ? (2)

[P.T.O.]

OR

Draw the molecular orbital diagram of dioxygen and calculate bond order. (2)

Q.10:(i) Draw the structure of diborane .

(ii) PbCl_4 is less stable than SnCl_4 but PbCl_2 is more stable than SnCl_2 . Why ? (1+1)

Q.11:(i) CO_2 is a gas while SiO_2 is solid at room temperature. Why?

(ii) SiCl_4 can be easily hydrolysed but CCl_4 does not hydrolysed. Why ?

(iii) Silicon shows a higher covalency than carbon. Why ? (1+1+1)

Q.12: An electron beam is accelerated by a potential difference of 10000 volts. What is the wavelength of the wave associated with the electron beam ?

(mass of electron = 9.1×10^{-31} Kg,

Charge of electron = 1.6×10^{-19} C)

OR

Calculate the uncertainty in the velocity of a cricket ball of mass 150 g , if the uncertainty in its position is of the order of $1\text{\AA}$.

($h = 6.6 \times 10^{-34}$ Kg m² s⁻¹) (3)

Q.13:(i) Draw the orbital structure of ethane .

(ii) Out of H_2O and H_2S which have high boiling point and why?

(iii) He_2 molecule does not exist. why ? (1+1+1)

Q.14:(i) Define " Charles law ". (1)

(ii) Calculate the temperature of 4 mol of gas occupying 5 dm³

[P.T.O.]

at 3.32 bar pressure.

$$(R = 0.083 \text{ bar dm}^3 / \text{mol/k}) \quad (2)$$

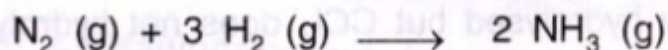
Q.15: Define the following terms :

(i) Hess's law

(ii) Standard enthalpy of atomization

(iii) Entropy (1+1+1)

Q.16: For the reaction :



$$\Delta H = -95.4 \text{ kJ} ; \quad \Delta S = -198.3 \text{ J/K}$$

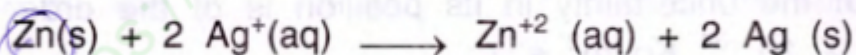
Calculate the temperature at which Gibb's free energy change is equal to zero . Predict the nature of the reaction at this temp. and above it . (3)

Q.17:(i) Given the standard electrode potentials :

$$\text{K}^+/\text{K} = -2.93 \text{ V} , \text{Ag}^+/\text{Ag} = +0.80 \text{ V} , \text{Cr}^{3+}/\text{Cr} = -0.74 \text{ V}$$

Out of these electrode which will be the strongest reducing agent? (1)

(ii) Represent the Galvanic cell in which the reaction takes place:



(a) Which of the electrode is negatively charged ?

(b) What are the carriers of the current in the cell ? (1+1)

Q.18: Chlorophyll, the green colouring material of plants contains 2.68 % of magnesium by mass. Calculate the number of moles of magnesium and atoms in 5.00 g of this complex. (Atomic mass of Mg=24) (2)

[P.T.O.]

- (i) Write the name and formula of the liquid substance.
- (ii) Why this compound is stored in dark coloured bottles ?
- (iii) How is the strength of the substance generally expressed ?
- (iv) What values are associated with the chemistry teacher ?

(1+1+1+1)

Q.24(i) State Le-chateliers principle. (1)

- (ii) Write conjugate base for the acids :



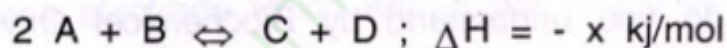
- (iii) What is the difference between solubility product and ionic product ? (1)

- (iv) Calculate the PH of of a solution having $[\text{H}_3\text{O}^+]$ of 10^{-3} . (2)

OR

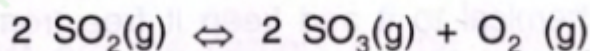
- (i) State " Common ion effect ". (1)

- (ii) For a hypothetical reaction :



What will be the effect on the equilibrium with

- (a) Decrease of temperature
- (b) addition of Helium gas (1+1)
- (iii) At 700 K , the equilibrium constant K_p , for the reaction



is $1.8 \times 10^{-3} \text{ k Pa}$. What is the numerical value in moles per litre of K_c for this reaction at this temperature ? (2)

[P.T.O.]

Q.25:(a) Account for the followings : (1+1+1)

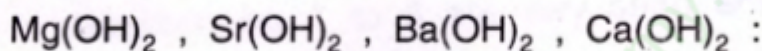
- (i) Be and Mg do not give flame colouration .
- (ii) Li is the strongest reducing agent.
- (iii) Potassium carbonate can not be prepared by Solvay process

(b) In what ways Li shows similarities to Mg in its chemical behavior. (any two) (2)

OR

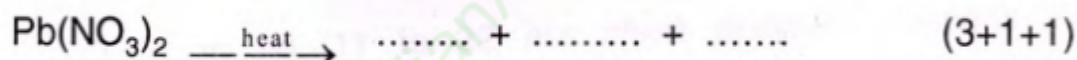
(a) Write three properties of Lithium which differ from the rest of the members of group 1.

(b) Arrange the following in order of the property mentioned :



increasing basic character

(c) Complete the reaction :



Q.26:(a) Explain the following reactions :

(i) Wurtz reaction

(ii) Friedel crafts alkylation (1+1)

(b) Convert :

(i) 1-bromopropane to propene

(ii) Sodium acetate to methane (1+1)

(c) Melting point of cis-2-butene is lower than that of trans-2-butene. why ? (1)

OR

[P.T.O.]

SESSION ENDING EXAMINATION – (2014-15)

Chemistry

CBSE CLASS – XI

Time: 3 Hrs. M.M: 70

General Instructions:

- (i) Question 1 to 5 one mark.
- (ii) Question 6 to 10 each two mark.
- (iii) Question 11 to 22 each three mark.
- (iv) Question 23 is value based question and carry four marks.
- (v) Questions 24 to 26 each five mark.

Q1 What do you understand by "Limiting reagent".

Q.2 State Hiesenberg Uncertainty Principle.

Q.3 The value of Vander was constants a and b are as given for two gases:

Gases	$a(\text{atm L}^2 \text{ mol}^{-1})$	$b(\text{mol}^{-1})$
C02	3.6	0.043
S02	6.7	0.056

Out of these = two gases which gas molecules will possess largest magnitude to intermolecular forces of attraction?

Q.4: In Lassaigne's test for detection of nitrogen in an organic compound the blue colour appears due to the formation.

Q.5: For a reaction both enthalpy change and entropy change are positive. Under what conditions the reaction will be spontaneous?

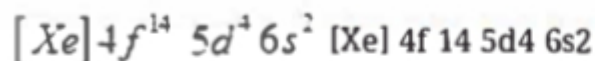
Q.6 (i) How many subshells are associated with $n=4$?

(ii) How many electron will be present in the subshells having m_z value of $-1/2$ for $n=4$?

Q.7 (i) Define electron gain enthalpy.

(ii) Why is the electron gain enthalpy of chlorine more negative than fluorine?

Q.8 (i). In what group of the periodic table the element will be found having electronic configuration :



(ii) Why is the first ionization enthalpy of nitrogen more than oxygen?

Q.9: Which out of NH_3 and NF_3 has higher dipole moment and why?

OR

Draw the molecular orbital diagram of dioxygen and calculate bond order.

Q.10 (i) Draw the structure of diborane.

(ii) PbCl_4 is less stable than SnCl_4 but PbCl_2 is more stable than SnCl_2 . Why?

Q.11 (i) CO_2 is a gas while SiO_2 is solid at room temperature. Why?

(ii) SiCl_4 can be easily hydrolysed but CCl_4 does not hydrolyse. Why?

(iii) Silicon shows a higher covalency than carbon. Why?

Q.12 An electron beam is accelerated by a potential difference of 10000 volts. What is the wavelength of the wave associated with the electron beam?

(mass of electron = $9.1 \times 10^{-31} \text{ kg}$, Charge of electrons = $1.6 \times 10^{-19} \text{ C}$)

OR

Calculate the uncertainty in the velocity of a cricket ball of mass 150 g, if the uncertainty in its position is of the order of 1 Å

($h = 6.6 \times 10^{-34} \text{ Kg m}^2 \text{ s}^{-1}$)

Q.13 (i) Draw the orbital structure of ethane.

(ii) Out of H_2O and H_2S which have high boiling point and why?

(iii) He_2 molecule does not exist. Why?

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(ii) Calculate the temperature of 4 mol of gas occupying 5 dm^3 at 3.32 bar pressure.

($R = 0.083 \text{ bar dm}^3/\text{mol/K}$)

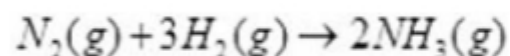
Q.15 Define the following terms:

(i) Hess's law

(ii) Standard enthalpy of atomization

(iii) Entropy

Q.16: For the reaction:



$$\Delta H = 95.4 \text{ kJ}; \Delta S = -198.3$$

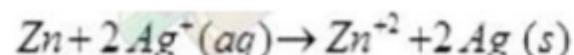
Calculate the temperature at which Gibbs's free energy change is equal to zero. Predict the nature of the reaction at this temp. and above it

Q17:(i) Given the standard electrode potentials

$$K^+/K = -2.93 \text{ V}, Ag^+/Ag = -0.80 \text{ V}, Cr^{3+}/Cr = -0.74 \text{ V}$$

Out of these electrode which will be the strongest reducing agent?

(ii) Represent the Galvanic cell in which the reaction takes place:



(a) Which of the electrode is negatively charged?

(b) What are the carriers of the current in the cell?

SAMPLE PAPER – I (SOLVED)

CHEMISTRY – XI

Time : 3 hrs.

M. M. : 70

General Instructions :

- (i) All questions are compulsory.
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- (iii) Question number 1 to 8 are very short answer questions carrying **one** mark each. Answer these in one word or about one sentence.
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- (v) Question number 19 to 27 are also short answer questions carrying **three** marks each. Answer them in about 40 words.
- (vi) Question number 28 to 30 are long answer questions carrying **Five** marks each. Answer them in about 70 words.
- (vii) Use log tables, if necessary. Use of calculator is not allowed.

1. The following set of quantum numbers is not possible. Explain, why ? 1
 $n = 2, l = 3, m = -3, m = +\frac{1}{2}$
2. Name a species which is isoelectronic with the following ion : 1
 N^{3-}
3. For an isolated system $\Delta U = 0$, what will be ΔS ? 1
4. Explain the structure of CO_2 in terms of resonance. 1
5. What are the conditions under which gases deviate from ideality. 1
6. Justify that the following reaction is a redox reaction : 1
 $H_2O (s) + F_2 (g) \rightarrow HF (g) + HOF (g)$
7. Write the IUPAC name of following : 1
 $CH_2 = CH - C \equiv CH$
8. What is the oxidation state of K in KO_2 . 1

9. Write balanced equation for reaction between : 1 × 2 = 2
 (a) Na_2O_2 and water
 (b) Ca metal with HCl

OR

Describe the hybridisation in case of PCl_5 . Why are the axial bonds longer as compared to equatorial bonds ? 2

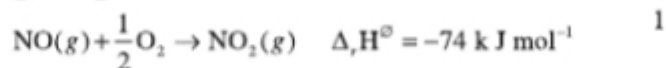
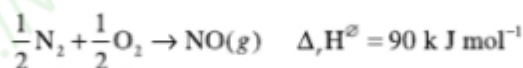
10. (a) Use molecular orbital theory to explain why Be_2 molecule does not exist. 1
 (b) Explain the formation of σ and π bonds in C_2H_4 with the help of diagram. Mention the hybrid state of two carbon atoms. 1
11. Calculate the mass of a photon with wavelength $3.6 \text{ \AA}$. 2
 ($h = 6.26 \times 10^{-34} \text{ JS}$)
12. Predict the products of electrolysis of an aqueous solution of AgNO_3 with silver electrodes. Write reaction that occurs at each electrode. 2
13. Write all the reactions involved in the preparation of sodium carbonate from sodium chloride in Solvay Process. 2
14. Complete the following equations : 2
 (a) $\text{Al} + \text{NaOH} + \text{H}_2\text{O} \rightarrow$
 (b) $\text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{A} \xrightarrow{\Delta} \text{B}$
15. A polluted water sample has been found to have 15 ppm CHCl_3 in it.
 (a) Express this value in percent by mass.
 (b) Determine the molality of chloroform in the water sample. 2
16. An alkene 'A' on ozonolysis gives a mixture of ethanal and pentan-3-one. Write the structure and IUPAC name of A. 2
17. What is the basic principle involved in the following process :
 (a) Partition chromatography 1
 (b) Distillation under reduced pressure 1
18. (a) Explain why Be and Mg do not give colour to the flame whereas other alkaline earth metals do. 1
 (b) Why alkali and alkaline earth metals cannot be prepared by chemical reduction methods ? 1

19. (a) Calculate the concentration of nitric acid in moles per litre in a sample which has a density 1.41 g mL^{-1} and the mass percent of HNO_3 in it being 69%. 2
- (b) How many atoms are present in 48u of C ? 1
20. (a) What are degenerate orbitals ? Give example. 1
- (b) Show that the circumference of the Bohr's orbit for H atom is an integral multiple of de Broglie's wavelength associated with the electron revolving around the orbit. 2
- OR**
- (a) Mention the difference between electromagnetic wave theory and Planck's quantum theory. 2
- (b) How many electrons can have quantum number values $n = 4, m_l = \frac{1}{2}$. 1
21. (a) Write the general outer electronic configuration of f-block elements. 1
- (b) Predict the formula of a compound formed between silicon and oxygen. 1
- (c) N has higher ionisation enthalpy than O and why? 1
22. (a) Calculate the total pressure in a mixture of 8 g O_2 and 4 g H_2 confined in a vessel of volume 1 dm^3 at 27°C . ($R = 0.083 \text{ bar dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$). 2
- (b) In terms of Charles's law explain why -273.15°C is the lowest possible temperature. 1
23. (a) The species H_2O and NH_3 can act both Bronsted acids and Bronsted Bases. For each give the corresponding conjugate acid and conjugate base. 2
- (b) What will be the pH of 0.002 M HCl ? 1
24. (a) Write the expression for K_p for the following reaction : 1
- $$\text{CH}_4(g) + \text{H}_2\text{O}(g) \rightleftharpoons \text{CO}(g) + 3\text{H}_2(g)$$
- (b) How will the values of K_p and composition of equilibrium mixture be affected by : 1
- (i) increasing the pressure
- (ii) using a catalyst

25. (a) Out of NH_3 , H_2O and HF which would you expect to have highest magnitude of hydrogen bonding and why ? 1
- (b) Write chemical equations to justify that H_2O_2 can act as an oxidising agent as well as a reducing agent. 2
26. Give reasons : 3
- (a) Why CO is a poisonous gas ?
- (b) Lead(IV) chloride is highly unstable towards heat.
- (c) Boric acid is not protic acid.
27. Explain the following terms : 3
- (a) Ozone hole
- (b) BOD
- (c) Green chemistry
28. (a) Derive the relationship between C_p and C_v for an ideal gas. 3
- (b) Given $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ $\Delta H = -92.4 \text{ kJ/mol}$.
What is the standard enthalpy of formation of NH_3 gas ?
- (c) The equilibrium constant for a reaction is 10. What will be sign of ΔG ?
Will this reaction be spontaneous ?

OR

- (a) Compare the thermodynamic stabilities of NO and NO_2 from the following data :



- (b) Write the Born Haber Cycle for the formation of $\text{CaCl}_2(\text{s})$. 2
- (c) For the reaction at 298 K : $2\text{A} + \text{B} \rightarrow \text{C}$

$$\Delta H = 400 \text{ kJ mol}^{-1} \text{ and } \Delta S = 0.2 \text{ kJ K}^{-1} \text{ mol}^{-1}$$

Determine the temperatures at which the reaction would be spontaneous.

SAMPLE PAPER – I (SOLVED)

CHEMISTRY – XI

Time : 3 hrs.

M. M. : 70

General Instructions :

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2. Name a species which is isoelectronic with the following ion : 1
 N^{3-}
3. For an isolated system $\Delta U = 0$, what will be ΔS ? 1
4. Explain the structure of CO_2 in terms of resonance. 1
5. What are the conditions under which gases deviate from ideality. 1
6. Justify that the following reaction is a redox reaction : 1
 $H_2O (s) + F_2 (g) \rightarrow HF (g) + HOF (g)$
7. Write the IUPAC name of following : 1
 $CH_2 = CH - C \equiv CH$
8. What is the oxidation state of K in KO_2 . 1

9. Write balanced equation for reaction between : 1 × 2 = 2
 (a) Na_2O_2 and water
 (b) Ca metal with HCl

OR

Describe the hybridisation in case of PCl_5 . Why are the axial bonds longer as compared to equatorial bonds ? 2

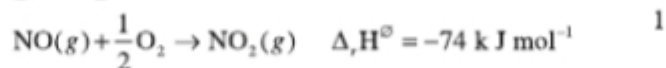
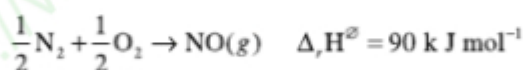
10. (a) Use molecular orbital theory to explain why Be_2 molecule does not exist. 1
 (b) Explain the formation of σ and π bonds in C_2H_4 with the help of diagram. Mention the hybrid state of two carbon atoms. 1
11. Calculate the mass of a photon with wavelength $3.6 \text{ \AA}$. 2
 ($h = 6.26 \times 10^{-34} \text{ JS}$)
12. Predict the products of electrolysis of an aqueous solution of AgNO_3 with silver electrodes. Write reaction that occurs at each electrode. 2
13. Write all the reactions involved in the preparation of sodium carbonate from sodium chloride in Solvay Process. 2
14. Complete the following equations : 2
 (a) $\text{Al} + \text{NaOH} + \text{H}_2\text{O} \rightarrow$
 (b) $\text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{A} \xrightarrow{\Delta} \text{B}$
15. A polluted water sample has been found to have 15 ppm CHCl_3 in it.
 (a) Express this value in percent by mass.
 (b) Determine the molality of chloroform in the water sample. 2
16. An alkene 'A' on ozonolysis gives a mixture of ethanal and pentan-3-one. Write the structure and IUPAC name of A. 2
17. What is the basic principle involved in the following process :
 (a) Partition chromatography 1
 (b) Distillation under reduced pressure 1
18. (a) Explain why Be and Mg do not give colour to the flame whereas other alkaline earth metals do. 1
 (b) Why alkali and alkaline earth metals cannot be prepared by chemical reduction methods ? 1

19. (a) Calculate the concentration of nitric acid in moles per litre in a sample which has a density 1.41 g mL^{-1} and the mass percent of HNO_3 in it being 69%. 2
- (b) How many atoms are present in 48u of C ? 1
20. (a) What are degenerate orbitals ? Give example. 1
- (b) Show that the circumference of the Bohr's orbit for H atom is an integral multiple of de Broglie's wavelength associated with the electron revolving around the orbit. 2
- OR**
- (a) Mention the difference between electromagnetic wave theory and Planck's quantum theory. 2
- (b) How many electrons can have quantum number values $n = 4, m_s = \frac{1}{2}$. 1
21. (a) Write the general outer electronic configuration of *f*-block elements. 1
- (b) Predict the formula of a compound formed between silicon and oxygen. 1
- (c) N has higher ionisation enthalpy than O and why? 1
22. (a) Calculate the total pressure in a mixture of 8 g O_2 and 4 g H_2 confined in a vessel of volume 1 dm^3 at 27°C . ($R = 0.083 \text{ bar dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$). 2
- (b) In terms of Charles's law explain why -273.15°C is the lowest possible temperature. 1
23. (a) The species H_2O and NH_3 can act both Bronsted acids and Bronsted Bases. For each give the corresponding conjugate acid and conjugate base. 2
- (b) What will be the pH of 0.002 M HCl ? 1
24. (a) Write the expression for K_p for the following reaction : 1
- $$\text{CH}_4(g) + \text{H}_2\text{O}(g) \rightleftharpoons \text{CO}(g) + 3\text{H}_2(g)$$
- (b) How will the values of K_p and composition of equilibrium mixture be affected by : 1
- (i) increasing the pressure
- (ii) using a catalyst

25. (a) Out of NH_3 , H_2O and HF which would you expect to have highest magnitude of hydrogen bonding and why ? 1
- (b) Write chemical equations to justify that H_2O_2 can act as an oxidising agent as well as a reducing agent. 2
26. Give reasons : 3
- (a) Why CO is a poisonous gas ?
- (b) Lead(IV) chloride is highly unstable towards heat.
- (c) Boric acid is not protic acid.
27. Explain the following terms : 3
- (a) Ozone hole
- (b) BOD
- (c) Green chemistry
28. (a) Derive the relationship between C_p and C_v for an ideal gas. 3
- (b) Given $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ $\Delta H = -92.4 \text{ kJ/mol}$.
What is the standard enthalpy of formation of NH_3 gas ?
- (c) The equilibrium constant for a reaction is 10. What will be sign of ΔG ?
Will this reaction be spontaneous ?

OR

- (a) Compare the thermodynamic stabilities of NO and NO_2 from the following data :



- (b) Write the Born Haber Cycle for the formation of $\text{CaCl}_2(\text{s})$. 2
- (c) For the reaction at 298 K : $2\text{A} + \text{B} \rightarrow \text{C}$

$$\Delta H = 400 \text{ kJ mol}^{-1} \text{ and } \Delta S = 0.2 \text{ kJ K}^{-1} \text{ mol}^{-1}$$

Determine the temperatures at which the reaction would be spontaneous.

SAMPLE PAPER - I (Unsolved)

CHEMISTRY – XI

Time allowed : 3 hrs.

M. M. : 70

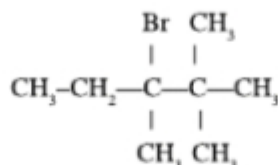
General Instructions :

- (i) All questions are compulsory.
- (ii) Marks for each question are indicated against it.
- (iii) Question number 1 to 8 are very short answer questions carrying one mark each. Answer these in one word or about one sentence.
- (iv) Question number 9 to 18 are short answer questions carrying 2 marks each. Answer these in about 30 words.
- (v) Question number 19 to 27 are also short answer questions carrying 3 marks each. Answer them in about 40 words.
- (vi) Question number 28 to 30 are long answer questions carrying 5 marks each. Answer them in about 70 words.
- (vii) Use log tables, if necessary. Use of calculator is not allowed.

1. Write the electronic configuration of Cu^+ ($Z = 29$). 1
2. State modern periodic law. 1
3. Give one example of intensive property. 1
4. What would you observe when HCl gas is passed into the saturated solution of NaCl ? 1
5. Calculate oxidation state of Mn in K_2MnO_4 . 1
6. Draw the structure of C_2H_5^+ . 1
7. Which reactive intermediate is formed when the covalent bond in $\text{CH}_3\text{--Li}$ under-goes hetrolytic cleavage ? 1

8. Predict the alkene as a major product obtained by the dehydrohalogenation of :

1



9. A sample of NaOH weighing 0.38 g is dissolved in water and solution is made 50.0 mL in a volumetric flask. Calculate the molarity of the resulting solution. (Molar mass of NaOH = 40 g mol⁻¹) 2
10. Electromagnetic radiation of wavelength of 242 nm is just sufficient to ionise sodium atom. Calculate the ionization enthalpy of sodium in kJ/mol. 2
11. Draw the energy level diagram of molecular orbitals in N₂ molecule. 2
12. On the basis of kinetic theory of gases explain why 2
- (a) gases exert pressure.
- (b) the volume of fixed amount of gas at constant pressure increases with the rise in temperature.
13. A student forgot to add the reaction mixture to a flask at 27° C and placed on the flame and heated it to 477° C. What fraction of air has been expelled out ? 2
14. State Le Chatelier principle. Predict the direction of the reaction : 2
- $$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$$
- when noble gas is added to the reaction mixture at constant pressure.
15. Balance the following chemical equation in alkaline medium : 2
- $$\text{I}^- + \text{MnO}_4^- \rightarrow \text{IO}_3^- + \text{MnO}_2$$
16. Contrast the action of heat on the following and explain your answer : 2
- (a) Na₂CO₃ and CaCO₃
- (b) Ca(NO₃)₂ and NaNO₃
17. Write the chemical equations for the following reactions : 2
- (a) Markownikov's rule (b) Friedel-Crafts reaction

OR

Explain the following technique of separation of organic compounds with suitable examples :

2

- (a) Chromatography (b) Differential extraction

18. (a) Why is fusion of organic compound with sodium metal is essential before performing tests for detection of extra elements in the organic compounds ?
(b) Write the chemistry involved in the test used for detection of nitrogen. 2
19. State Gay-Lussac's law combining volume of gas and Avogadro's law. Standard molar volume of any ideal gas is 22.7 L. Give the values of standard temperature and pressure. 3
20. (a) State Pauli's exclusion principle.
(b) Which rule will be disobeyed by the following electronic configurations : 3
(i) $2s^3$ (ii) $2s^2 2p_x^2 2p_y^1 2p_z^0$
21. Account for the following : 3
(a) Ionization enthalpy of nitrogen ($Z = 7$) is more than that of oxygen ($Z = 8$).
(b) Electron gain enthalpy of chlorine ($Z = 17$) is more negative than that of fluorine ($Z = 9$).
(c) Noble gases have high positive values of electron gain enthalpy.
22. (a) How many sigma (σ) and pi (π) bonds are there in $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$?
(b) Using VSEPR model predict the geometrical shapes of the following species : 3
(i) ClF_3 (ii) XeF_2
23. (a) If water molecules were linear instead of angular, would water (H_2O) be a good solvent as it is ? Why or why not ?
(b) How is H_2O_2 prepared by hydrated barium peroxide ? Show by chemical reactions that H_2O_2 acts both as oxidising and reducing agent. 3
24. Comment on each of the following observations :
(a) The mobilities of the alkali metal ions in aqueous solution are $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$.
(b) Lithium is only alkali metal to form a nitride directly.
(c) Alkaline earth metals are harder than alkali metals. 3

Sample Paper-II (Unsolved)

Subject : Chemistry

Time : 3 Hrs.

Class : IX

M.M. : 70

General Instructions :

- (i) All questions are compulsory.
- (ii) Marks for questions are indicated against each.
- (iii) Question no. 1-8 are very short answer questions carrying 1 mark each.
- (iv) Questions no. 9-18 are short answer questions carrying 2 marks each.
- (v) Question no. 19-27 are also short answer question carrying 3 marks each.
- (vi) Question no. 28-30 are long answer questions carrying 5 marks each.
- (vii) Answer should be brief and to the point.

1. Write the correct set of four quantum numbers for the valence electron of potassium ($Z = 19$). 1
2. Write the IUPAC name and symbol for the element with atomic number 199. 1
3. What property of molecules of real gases is indicated by van der Waals constant 'a'? 1
4. When 430J of work was done on a system, it lost 120J of energy as heat. Calculate the value of internal energy change (ΔU) for this process. 1
5. Write K_p in terms of K_c for the following chemical reaction : 1
$$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$$
6. Give an example of disproportionation reaction. 1
7. Name the isotope of hydrogen which contains equal number of protons and neutrons. 1
8. Write IUPAC name of the following organic compound : 1
$$\text{CH}_3 - \text{C}(\text{CH}_3)_2 - \text{CH} = \text{CH}_2$$
9. A sample of drinking water was found to be severely contaminated with chloroform (CHCl_3), supposed to be carcinogenic in nature. The level of contamination was 15ppm (by mass).
 - a) Express this in percent by mass.
 - b) Determine the molality of chloroform in the water sample.[Given molar mass of $\text{CHCl}_3 = 118.5\text{g}\cdot\text{mol}^{-1}$] 2

10. An electron has a speed of 40ms^{-1} accurate upto 99.99%. What is the uncertainty in locating its position? [Given $m_e = 9.11 \times 10^{-31}\text{ kg}$] 2
11. Give correct reason for the following : 2
- BF_3 has a zero dipole moment although the B-F bonds are polar.
 - All carbon to oxygen bonds in CO_3^{2-} are equivalent.
12. Balance the following redox reaction by ion electron method (in basic medium) : 2
- $$\text{MnO}_4^- (aq) + \Gamma^- (aq) \rightarrow \text{MnO}_2 (s) + \text{I}_2 (s)$$
13. Complete the following reactions : 2
- $\text{PbS} (s) + \text{H}_2\text{O}_2 (aq) \rightarrow$
 - $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow$

OR

Account for the following :

- Soft water lathers with soap but not hard water.
 - Temporary hardness of water can be removed by boiling.
14. a) Mention the type of hybridisation of each carbon in the compound $\text{CH}_3 - \text{CN}$. 2
- Draw the structure of 2-methylpropan-2-ol.
15. In sulphur estimation, 0.157g of an organic compound gave 0.4813g of barium sulphate. What is the percentage of sulphur in the compound? 2
16. Write the structures of products (A and B) of the following reactions : 2
- $\text{HC} \equiv \text{CH} \xrightarrow{\text{Na}} \text{A} \xrightarrow{\text{CH}_3\text{Br}} \text{B}$
 - $\text{BrH}_2\text{C} - \text{CH}_2\text{Br} \xrightarrow[\text{KOH}]{\text{Alcohol}} \text{A} \xrightarrow{\text{NaNH}_2} \text{B}$
17. Arrange the following in order of property mentioned against each : 2
- $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2, \text{BaCl}_2$ [Increasing ionic character]
 - $\text{Mg}(\text{OH})_2, \text{Ca}(\text{OH})_2, \text{Ba}(\text{OH})_2, \text{Sr}(\text{OH})_2$ [Increasing solubility in water]
18. Give reasons for the following :
- The enthalpy of neutralisation is always constant *i.e.*, 57.1 kJ/mol when a strong acid neutralises a strong base.
 - Neither q nor W is a state function but $q + W$ is a state function.

19. a) State Gay Lussac's law of gaseous volumes.
 b) What mass of CaCO_3 is required to react completely with 25ml of 0.75M HCl according to equation?

$$\text{CaCO}_3(s) + 2\text{HCl}(aq) \rightarrow \text{CaCl}_2(aq) + \text{CO}_2(g) + \text{H}_2\text{O}(l)$$
 [Given atomic masses : Ca = 40u, C = 12u, O = 16u, Cl = 35.5 u and H = 1 u]
20. Account for the following :
 a) Ionization enthalpy of nitrogen is more than that of oxygen.
 b) A cation is always smaller than its parent atom.
21. Calculate the standard enthalpy of formation of $\text{CH}_3\text{OH}(l)$ from the following data : 3

$$\text{CH}_3\text{OH}(l) + \frac{3}{2} \text{O}_2(g) \rightarrow \text{CO}(g) + 2\text{H}_2\text{O}(l) \quad ; \Delta_f H^\circ = 726 \text{ kJ/mol}$$

$$\text{C}(\text{graphite}) + \text{O}_2(g) \rightarrow \text{CO}_2(g) \quad ; \Delta_f H^\circ = 393 \text{ kJ/mol}^{-1}$$

$$\text{H}_2(g) + \frac{1}{2} \text{O}_2(g) \rightarrow \text{H}_2\text{O}(l) \quad ; \Delta_f H^\circ = 286 \text{ kJ/mol}$$
22. a) State Pauli's exclusion principle. 3
 b) Account for the following :
 i) Chromium has configuration $3d^5 4s^1$ and not $3d^4 4s^2$.
 ii) Bohr's orbit are called stationary states.
23. Explain the following terms : 3
 a) Photochemical smog
 b) Acid rain
 c) Eutrophication
24. a) Write the molecular orbital configuration of O_2^+ . Calculate its bond order and predict its magnetic behaviour.
 b) What is the state of hybridisation of nitrogen in NH_4^+ ion? 3
25. Account for the following : 3
 a) Potassium carbonate cannot be prepared by Solvay process.
 b) Beryllium and magnesium do not give colour to flame.
 c) Alkali metals and alkaline earth metals cannot be obtained by chemical reduction methods.

CBSE Board
Class XI Chemistry

Time: 3 Hours

Marks: 70

General Instructions

1. All questions are compulsory.
2. Question nos. 1 to 8 are very short answer type questions and carry 1 mark each.
3. Question nos. 9 to 18 are short answer type questions and carry 2 marks each.
4. Question nos. 19 to 27 are also short answer type questions and carry 3 marks each.
5. Question nos. 28 to 30 are long answer type questions and carry 5 marks each.
6. Use log tables if necessary, use of calculators is not allowed.

- Q. 1** Explain why o- nitrophenol has a lower boiling point than p – nitrophenol? [1]
- Q. 2** Out of CO_2 and BF_3 , which one of them will have a larger bond angle and why? [1]
- Q. 3** Which of the following will be a state function? [1]
- (i) Distance travelled in climbing the hill
 - (ii) Energy change in climbing the hill
- Q. 4** When sodium hydride is electrolyzed; hydrogen gas is liberated at which electrode? [1]
- Q. 5** Why are alkali metals used in photoelectric cells?
- Q. 6** Is the eclipsed conformation of propane has the same or different energy as the eclipsed conformation of ethane? [1]
- Q. 7** Which of the two - $\text{O}_2\text{NCH}_2\text{CH}_2\text{O}^-$ or $\text{CH}_3\text{CH}_2\text{O}^-$ is expected to be more stable and why? [1]
- Q. 8** Due to which compound, ozone depletion is caused in Antarctica? [1]
- Q. 9** Among the elements B, Al, C and Si: [2]
- (a) Which has the highest first ionization enthalpy?
 - (b) Which has the most negative electron gain enthalpy? Give reason.
- Q. 10** Which of the following statements related to the modern periodic table is incorrect and why? [2]
- (a) Each block contains a number of columns equal to the number of electrons that can occupy that sub shell.
 - (b) The d - block has 8 columns, because a maximum 8 electrons can occupy all the orbitals in d - sub shell.

OR

- (a) Write the atomic number of the element present in the third period and seventeenth group of the periodic table.
 - (b) Out of the elements Cr ($Z = 24$), Mg ($Z=12$) and Fe ($Z=26$), identify the element with five electrons in 3d sub shell.
- Q. 11** The drain cleaner contains small bits of aluminium which react with caustic soda to produce dihydrogen gas. What volume of dihydrogen at 20°C and one

- bar pressure will be released when 0.15 g of aluminium reacts. [2]
- Q. 12** Critical temperature of ammonia and carbon dioxide are 405.5 K and 304.10 K respectively. Which these gases will liquefy first when you start cooling from 500K to their critical temperature [2]
- Q. 13** Consider the reaction of water with F_2 and suggest, in terms of oxidation and reduction, which species are oxidized/ reduced. [2]
- Q. 14** An element 'A' belongs to group 2 of the periodic table. It shows anomalous behaviour from the rest of the elements of its group. It shows a diagonal relationship with another element 'B'. Chlorides of both 'A' and 'B' have bridged structure in vapour phase. Identify A and B and draw the structures of their respective chlorides. [2]
- Q. 15** A metal 'X' is present in chlorophyll. Identify the metal 'X'. How does this metal react with N_2 ? [2]
- Q. 16** Calculate the mass percent of different elements in sodium sulphate, (Na_2SO_4) [2]
- Q. 17** A compound (C_7H_{14}) on ozonolysis gives ethanal and pentan-3-one. What is the structure of alkene? [2]
- Q. 18** Why does the rain water normally have a pH of about 5.6? When does it become acid rain? [2]
- Q. 19** Calculate the molarity of a solution of ethanol in water in which the mole fraction of ethanol is 0.40. [3]
- Q. 20** Kavita was playing a game with her friends. As a part of the game they asked her to express a wish. She said that she wanted to be able to see the atom. Atomic dimensions are from 10^{-12} m and nucleus is 10^{-15} m; visible range in the electromagnetic spectrum is for wavelengths in the range of 10^{-7} m. As a student of chemistry [3]
- Describe how the world would look for kavita if she is granted her wish.
 - What value can you draw from this?
- Q. 21** (a) The 4f sub shell of an atom contains 12 electrons. What is the maximum number of electrons having the same spin in it? [3]
- Explain the meaning of $4p^6$.
 - Write the electronic configuration of the atom with atomic number
- OR**
- Calculate the total number of electrons present in one mole of methane.
 - An atomic orbital has $n = 3$. What are the possible values of l and m_l ?
- Q. 22** Explain the hybridisation of SF_4 ? [3]
- Q. 23** (a) Write the expression for equilibrium constant for the reaction: [3]
- $$H_2(g) + I_2(s) \rightleftharpoons 2HI(g)$$
- (b) Calculate the pH of a buffer solution containing 0.2 mole of NH_4Cl and 0.1 mole of NH_4OH per litre. Given K_b for $NH_4OH = 1.85 \times 10^{-5}$
- Q. 24** Consider the reaction: [3]

$2\text{SO}_2 \text{ g} + \text{O}_2 \text{ g} \rightleftharpoons 2\text{SO}_3 \text{ g} + 189.4 \text{ kJ}$. Indicate the direction in which the equilibrium with shift when:

- (a) Temperature is increased
- (b) Pressure is increased
- (c) Concentration of SO_2 is increase

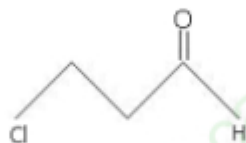
Q. 25 Balance $\text{P} + \text{HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2 + \text{H}_2\text{O}$ by oxidation number method. [3]

Q. 26 Write the IUPAC names of: [3]

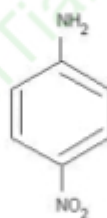
(a)



(b)



(c)



Q. 27 (a) Arrange the following carbanions in the increasing order of their stability:-



(b) What is the hybridisation of the negatively charged carbon atom in a carbanion?

Q. 28 (a) Compound 'A' with the molecular formula C_5H_8 reacts with hydrogen in [5]

the presence of Lindlar's catalyst to form a compound B with the molecular formula C_5H_{10} . A on reacting with sodium in liquid ammonia forms a compound 'C' with the same molecular formula as that of B. Identify 'A', 'B' and 'C'. Give the chemical reactions involved.

(b) Write the chemical reaction involved in Kolbe's electrolytic process. What are the products formed at cathode and anode?

OR

(a) Complete the reactions and identify A, B and C.



Roll No.	
Name	
Class & Section	

APEEJAY COMMON ANNUAL EXAMINATION, 2019-20

CHEMISTRY

Time Allowed : 3.00 Hrs.

Class – XI

Maximum Marks : 70

General Instructions :

- (a) *This question paper contains 37 Questions.*
- (b) *All questions are compulsory.*
- (c) *Section A contains 20 questions of one mark,
Section B contains 7 questions of two marks,
Section C contains 7 questions of three marks,
Section D contains 3 questions of five marks each.*
- (d) *Section A comprises of MCQs, Assertion-Reason Questions and Very Short Answer Questions. These questions are to be answered in one word, one sentence or as per the exact requirement of the question.*
- (e) *There is no overall choice. However, an internal choice has been provided in 2 questions of one mark, 2 questions of two marks, 4 questions of three marks and all the 3 questions of five marks.*
- (f) *Use of calculators is not permitted. You may use logarithmic tables if required.*

Section-A

Read the given passage and answer the questions 1 to 5 that follow :

The periodic table houses a few families and group of elements, each having its own particular properties. Early p-block comprises of group 13 and 14. Group 13 consists of B, Al, Ga, In, Tl with general configuration ns^2np^1 . Group 14 is Carbon family and comprises of C, Si, Ge, Sn, Pb with general configuration ns^2np^2 . Boron and carbon family form versatile compounds. Boric acid is used as an antiseptic for eyes whereas zeolites are shape selective catalysts.

1. What happens when boric acid is heated above 370K? 1
2. Name a zeolite which is used to convert alcohols directly into gasoline. 1
3. Draw the structure of Diborane. 1
4. What happens when diborane reacts with ammonia? 1
5. Why is PbCl_2 more stable than PbCl_4 ? 1

Questions 6 to 10 are one word answers :

6. 1 litre of a gas weighs 1.964 g at STP. What is its molecular mass? 1
7. Amongst ClF_3 or BCl_3 , which molecule has bond angle 120° . 1

OR

Based on VSEPR theory, total number of 90° bond angles present in the molecule of SF_4 are

8. The Vander waal's constant 'a' for different gases are given below : 1

Gas	a (atm L ⁻² mol ⁻² .)
O_2	1.36
N_2	1.39
CH_4	2.25
NH_3	4.17

Which gas can be most easily liquified?

9. Water is a liquid whereas H_2S is a gas. Give reason. 1
10. Algae reduce dissolved oxygen in water. This process is called 1

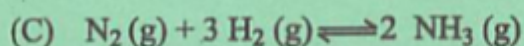
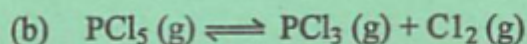
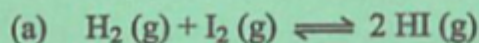
Questions 11 to 15 are multiple choice questions :

11. Which of the following pairs of d orbitals have electron density along the axis? 1
 - (a) d_{x^2} , d_{xz}
 - (b) d_{yz} , d_{xz}
 - (c) d_{x^2} , $d_{x^2-y^2}$
 - (d) d_{zy} , $d_{x^2-y^2}$
12. In second period of periodic table, first ionization enthalpy follows the order. 1
 - (a) $\text{Ne} > \text{F} > \text{O} > \text{N} > \text{C} > \text{Be} > \text{B} > \text{Li}$
 - (b) $\text{Ne} > \text{F} > \text{N} > \text{C} > \text{O} > \text{Be} > \text{B} > \text{Li}$

(c) $\text{Li} > \text{B} > \text{Be} > \text{C} > \text{O} > \text{N} > \text{F} > \text{Ne}$

(d) $\text{Ne} > \text{F} > \text{N} > \text{O} > \text{C} > \text{Be} > \text{B} > \text{Li}$

13. In which of the following reactions, the equilibrium remains unaffected on addition of small amount of argon at constant volume? 1



(d) The equilibrium will remain unaffected in all the three cases.

14. Standard electrode potential of three metals A, B and C are -1.3 V , $+0.8\text{ V}$ and -2.8 V respectively. The reducing power of these metals will be- 1

(a) $\text{A} > \text{B} > \text{C}$

(b) $\text{C} > \text{A} > \text{B}$

(c) $\text{C} > \text{B} > \text{A}$

(d) $\text{B} > \text{C} > \text{A}$

15. CO_2 is released most readily on heating which of the following? 1

(a) MgCO_3

(b) CaCO_3

(c) K_2CO_3

(d) Na_2CO_3

OR

Which of the following has lowest melting point?

(a) CaCl_2

(b) CaBr_2

(c) CaI_2

(d) CaF_2

Questions 16 to 20

(a) Both assertion and reason are correct statements and reason is the correct explanation of the assertion.

(b) Both assertion and reason are correct statements, but reason is not correct explanation of the assertion.

(c) Assertion is correct, but reason is wrong statement.

(d) Assertion is wrong, but reason is correct statement.

16. **Assertion :** Liquids tend to have maximum number of molecules at their surface. 1

Reason : Small liquid drops have spherical shape.

17. **Assertion :** One atomic mass unit is defined as one twelfth of the mass of one carbon -12 atom.

Reason : Carbon-12 isotope is the most abundant isotope of carbon and has been chosen as standard. 1

18. **Assertion :** A liquid crystallizes into a solid and is accompanied by decrease in entropy. 1

Reason : In crystals, molecules organize in an ordered manner.

19. **Assertion :** Permanent hardness of water cannot be removed by treatment with washing soda. 1

Reason : Washing soda reacts with soluble magnesium and calcium sulphate to form insoluble carbonates.

20. **Assertion :** Among isomeric pentanes, neopentane has least boiling point. 1

Reason : With an increase in branching, boiling point decreases.

Section-B

21. In the estimation of sulphur by Carius method, 0.468 g of an organic compound gives 0.668g of barium sulphate. Find the percentage of sulphur in the given compound. 2

[At mass; Ba = 137 u, S = 32 u, O = 16 u]

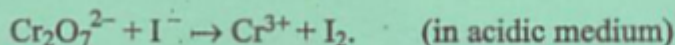
22. Calculate the energy associated with the first orbit of He^+ . What is the radius of this orbit? 2

OR

An electron has speed of 40 m sec^{-1} accurate upto 99.99%. What is uncertainty in locating its position? (Mass of electron = $9.1 \times 10^{-31} \text{ kg}$, velocity of electron = 300 ms^{-1} , $h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$)

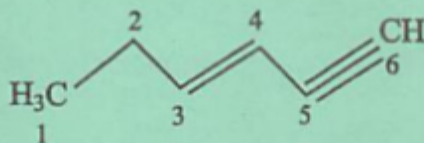
23. How is hydrogen peroxide prepared industrially? Explain why it is stored in wax-lined glass or plastic bottles? 2

24. Balance the equation : 2



25. What is BOD? If the value of BOD is low, what does it indicate? 2

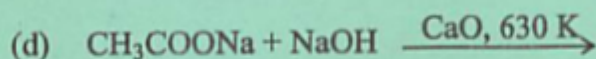
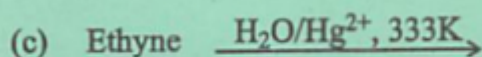
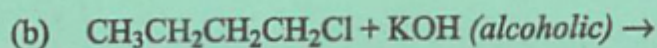
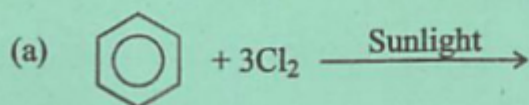
26. Write the IUPAC name of the following :



(a) Out of the 1st, 3rd and 5th carbon atom, which is most acidic? Why? 2

(b) How many sigma bonds are there in this compound?

27. Predict the products in the following reactions :



OR

Give complete balanced chemical equations for the following reactions : 2

(a) Wurtz reaction,

(b) Friedel Craft Acylation

Section-C

28. (a) Which is smaller in size- Fe²⁺ or Fe³⁺? Why? 3

(b) Write the general electronic configuration of f block elements.

(c) Write the IUPAC name and symbol of element with atomic number 117.

29. A compound on analysis found to contain following percentage composition Na = 43.4%, C = 11.3 %, and O = 45.3%. Determine the empirical and molecular formulae of this compound. The relative molecular mass of the compound is 106. (Atomic Mass of Na = 23, C = 12, O = 16) 3

OR

Commercially available Hydrochloric acid contains 38% HCl by mass.

(a) What is the molarity and molality of the solution if the density of the solution is 1.19 g/cc? 3

(b) What volume of concentrated HCl is needed to make 1 L of 0.2 M HCl solution?

30. Give the reactions involved in the Solvay's process for the preparation of sodium carbonate. 3

OR

- (a) Draw the structures of - 3
 (i) BeCl_2 (vapour) (ii) BeCl_2 (solid)

(b) Why are alkali metals strong reducing agents?

31. (a) An organic compound decomposes before attaining its boiling point. As this compound cannot be separated and purified by using simple distillation, suggest the method that can be used for its purification.

(b) What are free radicals? Discuss the stability of primary, secondary and tertiary free radicals. 3

32. (a) What will be the volume of hydrogen when 3 litres of it are cooled from 15°C to -73°C at constant pressure. 3

(b) What is Critical temperature?

OR

(a) Compare graph of P vs. V for real and an ideal gas.

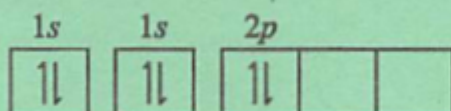
(b) What will be the pressure exerted by a mixture of 3.2g of methane and 4.4g of carbon dioxide at 9dm^3 flask at 27°C ? ($R=0.082\text{ L atm K}^{-1}\text{ mol}^{-1}$) 3

33. (a) Using s, p, d, f notations, describe the orbital with the following quantum number values :

(i) $n=4; l=0$ (ii) $n=3; l=2$

(b) Calculate de Broglie wavelength of an electron moving with 1% the speed of light.

(c) Which rule is violated in the following orbital diagram? 3



34. (a) Calculate the energy required to raise the temperature of 10 g of iron from 25°C to 500°C . (the specific heat capacity of iron is $0.45\text{ J }^\circ\text{C}^{-1}\text{ g}^{-1}$.)

(b) What mass of gold (specific heat capacity = $0.13\text{ J }^\circ\text{C}^{-1}\text{ g}^{-1}$) can be heated through the same temperature difference when supplied with the same amount of energy? 3

OR

(a) State the Third Law of Thermodynamics.

(b) Calculate and find whether it is possible to reduce MgO using carbon at 298 K.

